

AI Gesture Model Trainer Report

Project: Hand Gesture Drone Control System | Generated: 2026-08-10 21:30:25

1. Dataset Summary

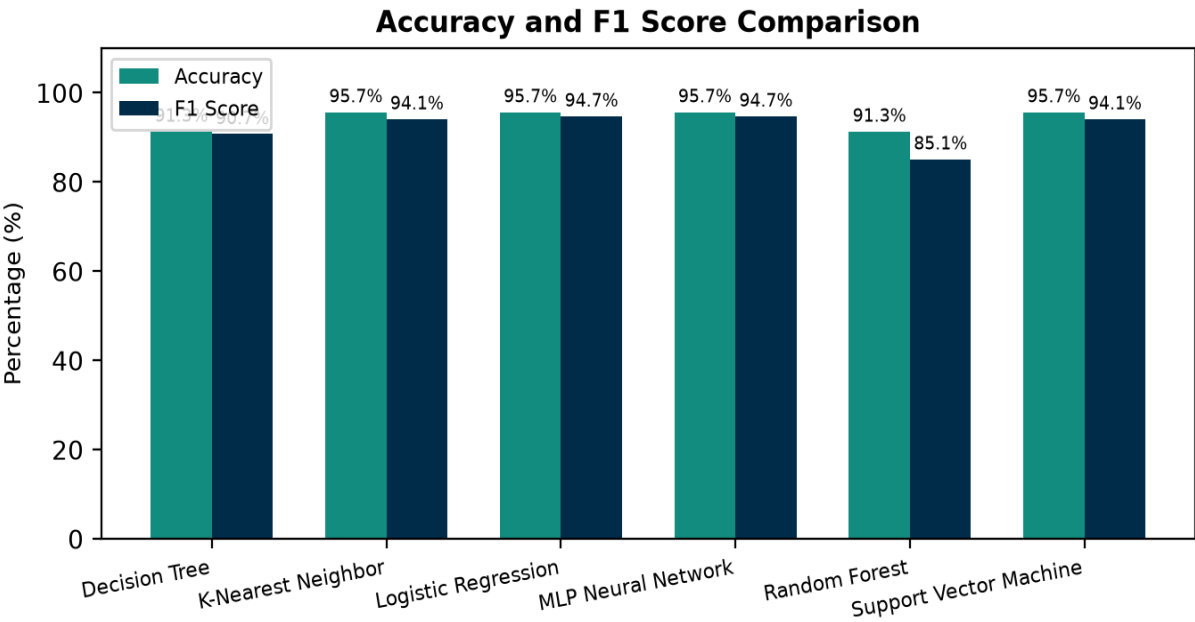
Metric	Value
Dataset Path	D:\university\SIC\AI\project_v2\AI_Drone_Edu_v2\dataset\test_dataset.csv
Total Sample Count	113
Feature Columns (Coordinates)	63
Number of Gesture Classes	9
Missing / NaN Values	0

2. Training Settings

Parameter	Value
Train / Test Ratio	80/20
Random State Seed	43
Cross Validation	5 Folds
Normalize Features	Yes (StandardScaler)
Shuffle Dataset	Yes

3. Model Performance Comparison

Model	Accuracy	Precision	Recall	F1	Train Time	Mem Size
Decision Tree	91.30%	90.74%	90.74%	90.74%	0.0022s	3.7 KB
K-Nearest Neighbor	95.65%	96.30%	94.44%	94.07%	0.0008s	45.7 KB
Logistic Regression	95.65%	97.22%	94.44%	94.71%	0.0105s	5.2 KB
MLP Neural Network	95.65%	97.22%	94.44%	94.71%	0.1105s	543.3 KB
Random Forest	91.30%	82.41%	88.89%	85.08%	0.1043s	335.1 KB
Support Vector Machine	95.65%	96.30%	94.44%	94.07%	0.0048s	44.5 KB



4. Selected Best Model Details: Logistic Regression

Metric	Result Score
Accuracy Score	95.65222%
Precision (Macro)	97.22222%
Recall (Macro)	94.44444%
F1 Score (Macro)	94.7090%
Single Sample Latency	0.0395 ms
Model Serialization Size	5.24 KB

Classification Report Detail:

	precision	recall	f1-score	support
Backward	1.00	1.00	1.00	3
Down	1.00	1.00	1.00	2
Forward	1.00	1.00	1.00	2
Hover	0.75	1.00	0.86	3
Land	1.00	1.00	1.00	3
Left	1.00	1.00	1.00	3
Right	1.00	1.00	1.00	2
Takeoff	1.00	1.00	1.00	3
Up	1.00	0.50	0.67	2
accuracy			0.96	23
macro avg	0.97	0.94	0.95	23
weighted avg	0.97	0.96	0.95	23